

A graded behavioural comparison framework for labelled Petri nets and qualitative network models

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Abstract

Given two finite executable labelled models and a declared observable interface, the comparison problem is to identify the strongest supported relation between their reachable transition systems without conflating local structure, linear traces and branching behaviour. We formulate a graded reporting framework whose exclusive output is strong bisimilarity, weak bisimilarity, mutual weak simulation, one-way weak simulation or non-comparability. Here “graded” denotes this implication-aware reporting priority, not an algebraic grading. For the implemented relation-deletion algorithm, we establish monotonicity, finite termination and greatest-fixed-point correctness. We also state and prove the relation hierarchy, invariance under the controlled silent edge subdivisions used by the benchmark, and the inability of label-blind structural comparators to characterize labelled behaviour in general. Six construction-level cases separate identity, silent refinement, label mismatch, order change, added behaviour and trace-equivalent branching. On the binary weak-equivalence task, weak bisimulation classified all 6 cases correctly, whereas truncated trace equality classified 5, labelled transition-system graphlet agreement 4, a structural profile 3 and direct PN-GDDA 2. An independent attacker-defender implementation agreed on all 67,600 ordered comparisons in the declared universe of one- and two-state systems, and Python and mCRL2 agreed on all 42 reported strong/weak decisions. Public GINsim models, HPN-DREAM/CASPOTS model families and curated Petri nets demonstrate cross-format use and expose dependence on update semantics and observational interfaces. These applications support a reproducible model-level audit; they do not establish predictive performance, cell-line specificity or organism-level equivalence.

Keywords: labelled Petri nets, labelled transition systems, weak bisimulation, simulation preorder, behavioural equivalence, formal verification, model checking, qualitative network models

2020 MSC: 68Q85, 68Q60, 68Q55.

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1. Introduction

Executable models are compared for several different purposes. An investigator may ask whether two graphs have similar local structure, whether they generate the same finite event sequences, whether they preserve branching choices, or whether the behaviour of one is contained in the other. These questions are not interchangeable. A structural score may ignore labels and initial states; trace equality records event order but not the point at which a choice is resolved; and an equivalence relation does not report asymmetric containment.

Let N_1 and N_2 be executable labelled models and let Σ be a declared observable alphabet. The problem considered here is

$$(N_1, N_2, \Sigma) \mapsto (\mathcal{L}(N_1), \mathcal{L}(N_2)) \mapsto C_{\Sigma}(\mathcal{L}(N_1), \mathcal{L}(N_2)), \quad (1.1)$$

where $\mathcal{L}(N_i)$ is the finite reachable labelled transition system and C_{Σ} returns the strongest supported class among strong bisimilarity, weak bisimilarity, mutual weak simulation, either one-way weak simulation

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and non-comparability. The word *graded* refers to this ordered reporting rule: relations implied by a stronger result are not reported as separate answers. It does not denote a graded algebraic structure.

Labelled place/transition nets provide an operational model of causality, conflict and concurrency [19]. Their reachable markings induce labelled transition systems, so standard behavioural relations from concurrency theory and model checking can be applied [18, 2, 26]. Internal actions are represented by τ . Strong bisimulation matches them as ordinary actions; weak bisimulation and weak simulation abstract finite internal paths. The resulting distinction is useful whenever two model descriptions expose the same interface at different internal resolutions.

No single structural or linear-time statistic solves the problem in (1.1). Graphlet methods make local structural comparison more sensitive than elementary size summaries. PN-GDDA, for example, compares distributions over 151 directed bipartite Petri-net graphlets and 592 published orbit slots [27]. It intentionally answers a structural question. In contrast, weak-trace equality is label-aware and global in event order but forgets branching. The framework therefore keeps exact behavioural relations separate from diagnostic scores and states which information each output uses.

The formal layer developed below makes the observational interface, simulation orientation and greatest-fixed-point operator explicit. Its properties justify the priority used by C_Σ and show why controlled silent subdivision preserves weak, but generally not strong, behaviour. Minimal counterexamples separate equal traces from bisimulation and label-blind structure from labelled semantics. The implementation is then tested against construction-level ground truth, mCRL2 and a separately coded attacker-defender oracle. The latter is exhaustive within the declared finite universe of all one- and two-state systems over $\{a, \tau\}$.

Qualitative biological networks provide heterogeneous applications of the formal problem. Petri nets have long been used for biochemical and regulatory systems [16, 5, 14], while executable cell biology requires explicit operational assumptions [25, 10]. Boolean models add a further semantic choice: synchronous, unitary-asynchronous and most-permissive updates can expose different reachable behaviour [21]. Model checking has also been applied to Boolean-network control, while formal constraints support logical-model inference [7, 3]. We therefore use externally authored GINsim models to test cross-format execution, HPN-DREAM/CASPOTS families to confront formal classes with held-out measurements [15, 24], and compact curated Petri nets to examine interface-dependent interpretation. These are applications of the comparison machinery, not premises of its correctness.

The contributions are:

1. a formal classifier for two finite labelled models under a declared observational interface;
2. fixed-point correctness and termination results for the implemented relation-deletion algorithms, together with the hierarchy that supports the reporting rule;
3. controlled examples and propositions that separate silent refinement, label-blind structure, finite traces, branching and directional simulation;
4. a synthetic construction-level benchmark evaluated against trace, structural-profile, LTS-GDA and direct PN-GDDA comparators;
5. independent checking with mCRL2 and exhaustive agreement with a dual simulation-game oracle on 67,600 ordered small-LTS pairs;
6. cross-format applications to public GINsim and HPN-DREAM/CASPOTS models and to curated Petri nets, with update-semantics sensitivity and held-out data kept distinct from formal correctness; and
7. an end-to-end reproducibility package containing source models, hashes, tests, an executed notebook and generated outputs.

Table 1 separates the information represented by each comparator. The integrated audit is the only evaluated path that combines local structure, global event order, branching, abstraction of hidden events, directional containment and a held-out data check.

Table 1: Positioning of the evaluated comparison layers. “Local” means that order is represented only inside a bounded motif rather than recursively over successor states

Method	Structure	Global order	Branching	τ abstraction	Directional relation	Held-out data
PN-GDDA [27]	yes	no	no	no	no	no
Structural profile	yes	no	no	no	no	no
Trace equality	no	yes	no	yes	no	no
LTS-GDA	yes/local	no	no	no	no	yes
Integrated behavioural audit	yes	yes	yes	yes	yes	yes

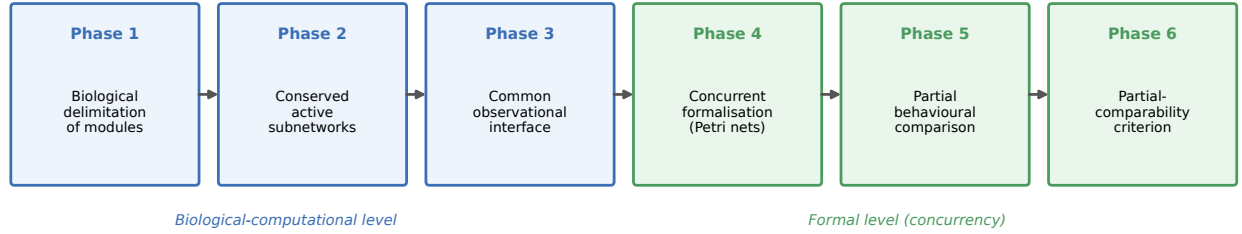


Figure 1: Six-stage workflow. Stages 1–3 delimit the model-comparison question and observational interface; stages 4–6 construct executable Petri nets, compare their reachable behaviour and assign a model-level relation

The operational sequence is summarised in Fig. 1. Biological delimitation and interface declaration precede formal encoding, reachability construction and the graded decision rule. This separation makes the modelling assumptions visible before any equivalence calculation is performed.

2. Mathematical framework

2.1. Labelled Petri nets and induced transition systems

Definition 2.1 (Labelled place/transition net). A labelled place/transition net is a tuple

$$N = (P, T, \text{pre}, \text{post}, \ell, m_0), \quad (2.1)$$

where P and T are finite sets of places and transitions, $\text{pre}, \text{post} : T \rightarrow \mathbb{N}^P$ are the input and output multisets, $m_0 \in \mathbb{N}^P$ is the initial marking, and $\ell : T \rightarrow \Sigma \cup \{\tau\}$ assigns either an observable action in the declared interface Σ or the internal action τ .

Under standard interleaving semantics, t is enabled at m when $\text{pre}(t) \leq m$ componentwise; firing it changes m to $m - \text{pre}(t) + \text{post}(t)$. The implementation uses a worklist to enumerate reachable markings explicitly. A state cap and a token bound make failure to establish a finite state space explicit rather than silently truncating the relation calculation. All curated Petri nets used in Section 4 are safe, but the definitions below apply to every finite reachable system.

Definition 2.2 (Finite labelled transition system). A finite labelled transition system (LTS) is

$$L = (S, \Sigma_\tau, \longrightarrow, s_0), \quad \Sigma_\tau = \Sigma \cup \{\tau\}, \quad (2.2)$$

where S is a finite state set, $s_0 \in S$, and $\longrightarrow \subseteq S \times \Sigma_\tau \times S$. We write $s \xrightarrow{a} s'$ for $(s, a, s') \in \longrightarrow$. The reachable semantics of the net in Definition 2.1 is denoted by $\mathcal{L}(N)$.

2.2. Observable and weak transitions

Definition 2.3 (Silent closure and weak target sets). For $s \in S$, let

$$\text{Cl}_\tau(s) = \{u \in S : s \xrightarrow{\tau^*} u\}. \quad (2.3)$$

For $a \in \Sigma$, define

$$W_a(s) = \{v : \exists u, w \quad s \xrightarrow{\tau^*} u \xrightarrow{a} w \xrightarrow{\tau^*} v\}, \quad W_\tau(s) = \text{Cl}_\tau(s). \quad (2.4)$$

We write $s \xRightarrow{a} v$ when $v \in W_a(s)$.

The convention $W_\tau = \text{Cl}_\tau$ is important: a direct silent move on one side may be matched by zero or more silent moves on the other. It is also the convention implemented by the Python engine and used in the AUT exports checked by mCRL2.

2.3. Simulation and bisimulation

Let $L_i = (S_i, \Sigma_\tau, \longrightarrow_i, s_i^0)$ for $i \in \{1, 2\}$.

Definition 2.4 (Weak simulation). A relation $R \subseteq S_1 \times S_2$ is a weak simulation from L_1 to L_2 if, whenever $(x, y) \in R$ and $x \xrightarrow{a}_1 x'$, there is $y' \in W_a^2(y)$ such that $(x', y') \in R$. We write

$$L_1 \preceq_w L_2 \quad (2.5)$$

if (s_1^0, s_2^0) belongs to some such relation. Thus the right-hand system L_2 simulates the left-hand system L_1 ; the left-hand behaviour is contained in the right-hand behaviour under the interface.

Definition 2.5 (Weak and strong bisimulation). A relation $R \subseteq S_1 \times S_2$ is a weak bisimulation if it satisfies the condition in Definition 2.4 and, for every $(x, y) \in R$ and $y \xrightarrow{a}_2 y'$, there is $x' \in W_a^1(x)$ with $(x', y') \in R$. The initial systems are weakly bisimilar, written $L_1 \sim_w L_2$, when their initial pair belongs to such a relation. Strong bisimulation, written $L_1 \sim_s L_2$, replaces every weak target set $W_a^i(x)$ by the direct target set $\{x' : x \xrightarrow{a}_i x'\}$, including for $a = \tau$.

These are standard branching-time relations [18, 26]. The explicit orientation in (2.5) prevents the common ambiguity between “simulates” and “is simulated by” in one-way results.

2.4. Weak trace language

Definition 2.6 (Exact and truncated weak traces). The finite weak observable language of L is

$$L_w(L) = \{a_1 \cdots a_r \in \Sigma^* : s_0 \xRightarrow{a_1} \cdots \xRightarrow{a_r} s \text{ for some } s \in S\}. \quad (2.6)$$

Its depth- k restriction is $L_{\leq k}(L) = \{u \in L_w(L) : |u| \leq k\}$. The diagnostic distance is

$$d_k(L_1, L_2) = 1 - \frac{|L_{\leq k}(L_1) \cap L_{\leq k}(L_2)|}{|L_{\leq k}(L_1) \cup L_{\leq k}(L_2)|}. \quad (2.7)$$

The implementation computes the prefix-closed truncated sets in Definition 2.6. The statement $d_k = 0$ is therefore not identified with exact weak-trace equivalence. Exact weak-trace results in this study come only from mCRL2 or from a finite construction for which the complete language is enumerated analytically.

118 2.5. Graded behavioural classification

Definition 2.7 (Graded classifier). For finite L_1, L_2 over the declared interface Σ , define $C_\Sigma(L_1, L_2)$ by the following priority:

$$C_\Sigma(L_1, L_2) = \begin{cases} \text{strong,} & L_1 \sim_s L_2, \\ \text{weak,} & L_1 \not\sim_s L_2 \text{ and } L_1 \sim_w L_2, \\ \text{mutual,} & L_1 \not\sim_w L_2, L_1 \preceq_w L_2, L_2 \preceq_w L_1, \\ \text{left_in_right,} & L_1 \preceq_w L_2, L_2 \not\preceq_w L_1, \\ \text{right_in_left,} & L_2 \preceq_w L_1, L_1 \not\preceq_w L_2, \\ \text{noncomparable,} & \text{otherwise.} \end{cases} \quad (2.8)$$

119 The output is a reporting class, not a newly proposed equivalence relation.

120 **Proposition 2.8** (Exclusive reporting). *The classifier in Definition 2.7 assigns exactly one class to every ordered*
 121 *pair of finite LTSs.*

122 *Proof.* The first two cases are selected before any simulation-only case. If neither is selected, the two
 123 simulation predicates have exactly four Boolean combinations: both true, only the first true, only the
 124 second true, or both false. These are the last four cases of (2.8), respectively. The cases are exhaustive and
 125 disjoint by their stated guards. $\square$

126 2.6. Fixed-point computation and formal properties

For $a \in \Sigma_\tau$, let $D_a^i(x) = \{x' : x \xrightarrow{a}_i x'\}$ and $U = S_1 \times S_2$. Define operators on $\mathcal{P}(U)$ by

$$\Phi_{\preceq}(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in W_a^2(y), (x', y') \in R\}, \quad (2.9)$$

$$\Phi_w(R) = \{(x, y) \in \Phi_{\preceq}(R) : \forall a, y' \in D_a^2(y) \exists x' \in W_a^1(x), (x', y') \in R\}, \quad (2.10)$$

$$\Phi_s(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in D_a^2(y), (x', y') \in R, \\ \forall a, y' \in D_a^2(y) \exists x' \in D_a^1(x), (x', y') \in R\}. \quad (2.11)$$

127 **Theorem 2.9** (Correctness of relation deletion). *For each $\Phi \in \{\Phi_{\preceq}, \Phi_w, \Phi_s\}$, the operator is monotone. The*
 128 *implemented algorithm starts with $R = U$, repeatedly deletes any currently examined pair $z \notin \Phi(R)$, and stops after*
 129 *a complete pass with no deletion. On finite LTSs it terminates and returns the greatest fixed point $\nu\Phi$. Consequently,*
 130 *testing membership of (s_1^0, s_2^0) computes, respectively, weak simulation in the orientation of Definition 2.4, weak*
 131 *bisimilarity, and strong bisimilarity.*

132 *Proof.* Let $R \subseteq Q \subseteq U$. Every witness pair required by (2.9)–(2.11) that belongs to R also belongs to Q .
 133 Hence $\Phi(R) \subseteq \Phi(Q)$, proving monotonicity for all three operators.

134 Each deletion strictly decreases the finite relation, so at most $|U|$ deletions can occur. A pass with no
 135 deletion therefore occurs after finitely many passes. By the Knaster–Tarski theorem on the finite complete
 136 lattice $\mathcal{P}(U)$, $G = \nu\Phi$ exists. Initially $G \subseteq U$. Suppose $G \subseteq R$ immediately before a deletion. If $z \in G$,
 137 then $z \in \Phi(G) \subseteq \Phi(R)$ by fixed-point equality and monotonicity, so z cannot be deleted. Thus $G \subseteq R$ is
 138 an invariant.

139 Let R_* be the relation at termination. Every retained pair passed its transfer test in the final no-
 140 deletion pass, hence $R_* \subseteq \Phi(R_*)$. Conversely, if $z \notin R_*$, it was deleted from some earlier relation R_z
 141 because $z \notin \Phi(R_z)$. Since $R_* \subseteq R_z$, monotonicity gives $\Phi(R_*) \subseteq \Phi(R_z)$, and therefore $z \notin \Phi(R_*)$. Thus
 142 $\Phi(R_*) \subseteq R_*$ and $R_* = \Phi(R_*)$.

143 The invariant gives $G \subseteq R_*$. Since R_* is a fixed point and G is the greatest fixed point, $R_* \subseteq G$; hence
 144 $R_* = G$. Finally, (2.9), (2.10) and (2.11) are exactly the transfer clauses in Definitions 2.4 and 2.5. $\square$

145 *Remark 2.10* (Resource bound). Writing $n_i = |S_i|$, the candidate relation uses $O(n_1 n_2)$ pairs. At most $n_1 n_2$
 146 deletions and $n_1 n_2 + 1$ full passes are possible. A pass examines the current pairs and their direct moves;
 147 matching scans the corresponding direct or weak target set. Silent closures are cached, while visible weak
 148 target sets are constructed on demand by the current code. For this reason the implementation claim
 149 is deliberately parameterised by the number of examined pairs, outgoing moves, weak-target sizes and
 150 refinement passes rather than replaced by an unjustified tight time bound.

Proposition 2.11 (Hierarchy of behavioural relations). *For finite LTSs under Definitions 2.3–2.6,*

$$L_1 \sim_s L_2 \implies L_1 \sim_w L_2 \implies (L_1 \preceq_w L_2 \wedge L_2 \preceq_w L_1) \implies L_w(L_1) = L_w(L_2). \quad (2.12)$$

151 *The converses do not hold in general.*

152 *Proof.* A direct matching move is a permitted weak matching move, so every strong bisimulation satisfies
 153 the weak clauses. Every weak bisimulation satisfies each one-sided simulation clause by Definition 2.5.

154 It remains to show the language consequence. If $L_1 \preceq_w L_2$, take a concrete path witnessing any word
 155 in $L_w(L_1)$. Apply the simulation clause to each direct transition of that path. Observable transitions are
 156 matched by $\tau^* \alpha \tau^*$ paths and silent transitions by τ^* paths, while successor pairs remain in the simulation
 157 relation. Erasing τ therefore gives the same observable word in L_2 . Induction on the finite concrete path
 158 yields $L_w(L_1) \subseteq L_w(L_2)$. Simulation in both directions gives equality. The counterexamples in Proposition
 159 2.12 and Example 2.14 refute the first and last converses. Mutual simulation without bisimulation is
 160 obtained, for example, by comparing a system with initial choices $a.(b + c)$ and $a.b$ against $a.(b + c)$:
 161 each simulates the other, but the extra $a.b$ successor cannot be related bisimilarly to the state offering
 162 both b and c . $\square$

Proposition 2.12 (Controlled silent edge refinement). *Let L contain $p \xrightarrow{\alpha} q$ with $\alpha \in \Sigma$ and $p \neq q$. Construct L' by copying all original states and transitions except this edge, and replacing it by*

$$\bar{p} \xrightarrow{\alpha} r \xrightarrow{\tau} \bar{q}, \quad (2.13)$$

163 *where r is fresh and has no other incident behaviour. Then $L \sim_w L'$. Repeated refinements of this form also preserve*
 164 *weak bisimilarity. If p and $\bar{p}$ are initial, the selected edge is the only α -edge from $\bar{p}$ in L' , and every α -successor of*
 165 *p in L has no outgoing τ transition, then $L \not\sim_s L'$.*

166 *Proof.* Let $R = \{(x, \bar{x}) : x \in S\} \cup \{(q, r)\}$. For every copied pair other than the source of the selected edge,
 167 direct transitions are copied and hence can be matched weakly. At $(p, \bar{p})$, the original α -move to q is
 168 matched by $\bar{p} \xrightarrow{\alpha} r \xrightarrow{\tau} \bar{q}$, ending in $(q, \bar{q})$. Conversely, $\bar{p} \xrightarrow{\alpha} r$ is matched by $p \xrightarrow{\alpha} q$, ending in (q, r) .

169 At (q, r) , each move of q is matched after the silent step from r to $\bar{q}$. The only direct move from r ,
 170 namely $r \xrightarrow{\tau} \bar{q}$, is matched by the zero-length silent path at q , ending in $(q, \bar{q})$. Thus R satisfies both
 171 weak transfer clauses. Repeating the construction preserves weak bisimilarity at each step, so the finite
 172 sequence follows by transitivity.

173 Under the stated additional conditions, a strong match for $\bar{p} \xrightarrow{\alpha} r$ must relate r to an α -successor x
 174 of p . The move $r \xrightarrow{\tau} \bar{q}$ would then require a direct τ -move from x , which is excluded. Hence no strong
 175 bisimulation contains the initial pair. $\square$

176 **Proposition 2.13** (Limit of label-blind structural comparison). *Let M be any comparator determined only by*
 177 *unlabelled structure and invariant under relabelling of transitions. Then M cannot characterize weak-trace equality,*
 178 *weak simulation or weak bisimilarity for labelled systems in general.*

179 *Proof.* Take two systems with the same two states and one directed edge. Label the edge a in L_a and $b \neq a$
 180 in L_b . Their unlabelled structures are identical, so relabelling invariance gives $M(L_a, L_b) = M(L_a, L_a)$.
 181 However, $L_w(L_a) = \{\epsilon, a\}$ and $L_w(L_b) = \{\epsilon, b\}$. Neither initial observable move can be matched by the
 182 other system, so there is no weak simulation in either direction and no weak bisimulation. Therefore the
 183 value assigned to a self-match by M cannot characterize any of these labelled relations. $\square$

Proposition 2.13 does not identify a defect in PN-GDDA. It states that a label-blind structural method answers a different question and cannot substitute for a labelled behavioural relation.

Example 2.14 (Equal traces, different branching). Let L_{late} have transitions

$$\ell_0 \xrightarrow{a} \ell_1, \quad \ell_1 \xrightarrow{b} \ell_b, \quad \ell_1 \xrightarrow{c} \ell_c, \quad (2.14)$$

and let L_{early} have

$$e_0 \xrightarrow{a} e_b, \quad e_0 \xrightarrow{a} e_c, \quad e_b \xrightarrow{b} e'_b, \quad e_c \xrightarrow{c} e'_c. \quad (2.15)$$

Both exact prefix-closed languages are $\{\epsilon, a, ab, ac\}$. Nevertheless, $L_{\text{late}} \not\sim_w L_{\text{early}}$. There are no silent moves, and matching either early a -successor with ℓ_1 leaves one of the two late actions unmatched. Directionally, $L_{\text{early}} \preceq_w L_{\text{late}}$: the late-choice state can match the selected b or c continuation of either early branch. The reverse $L_{\text{late}} \preceq_w L_{\text{early}}$ fails because no single early a -successor offers both continuations. Thus exact trace equality does not determine branching-time equivalence or simulation orientation.

2.7. Algorithms

Algorithm 2.15 (Reachable LTS construction). **Input:** labelled executable model N , initial state or marking s_0 , state cap B . **Output:** reachable finite LTS or an explicit cap error.

1. Initialize a worklist and state index with s_0 .
2. Remove one discovered state s and enumerate every enabled model move.
3. Execute each move, add its labelled edge, and index any new successor.
4. Abort if the token bound or B is exceeded; otherwise continue until the worklist is empty.

Algorithm 2.16 (Weak targets). **Input:** LTS L , state s , action a . **Output:** $W_a(s)$.

1. Compute and cache $Cl_\tau(s)$ by graph search.
2. If $a = \tau$, return this closure.
3. Otherwise inspect every a -edge from every state in the closure and union the cached silent closures of its targets.

Algorithm 2.17 (Greatest-fixed-point relation). **Input:** finite L_1, L_2 and transfer operator Φ . **Output:** whether the initial pair belongs to $\nu\Phi$.

1. Set $R \leftarrow S_1 \times S_2$.
2. Scan a snapshot of the current pairs. Delete (x, y) immediately whenever $(x, y) \notin \Phi(R)$.
3. Repeat Step 2 until an entire pass deletes no pair.
4. Return whether $(s_1^0, s_2^0) \in R$.

Algorithm 2.18 (Graded report). **Input:** finite L_1, L_2 . **Output:** one class in Definition 2.7 plus separate diagnostics.

1. Evaluate strong and weak bisimilarity and both ordered simulations.
2. Apply the priority in (2.8); stop at the first satisfied case.
3. Compute d_k and structural scores only as attached diagnostics.

Table 2 distinguishes exact relations from finite-depth or structural diagnostics.

3. Algorithms and computational validation

3.1. Synthetic benchmark with construction-level ground truth

Six deterministic cases were specified independently of the biological models. A 12-step labelled workflow is used for the first five cases. The identity case copies it exactly. Silent refinement subdivides selected observable edges with internal τ steps. Label mismatch replaces a recurring observable. Label order swap preserves topology and label counts while changing two consecutive events. Added branch retains the reference workflow and adds one observable outcome. The sixth case is the standard branching-time distinction $a.(b + c)$ versus $a.b + a.c$: the trace sets agree, but one model resolves its choice before the other. Table 3 records the relation expected from each construction.

Table 2: Inputs and outputs of the computational components

Component	Input	Output	Status
Reachability	labelled executable model	finite reachable LTS	exact under stated caps
Strong bisimulation	L_1, L_2	yes/no	formal
Weak bisimulation	L_1, L_2	yes/no	formal
Weak simulation	ordered L_1, L_2	yes/no	formal
Trace distance	L_1, L_2, k	d_k	truncated diagnostic
PN-GDDA	Petri-net structure	similarity score	structural diagnostic
LTS-GDA	reachable labelled graph	similarity score	local diagnostic

Table 3: Synthetic benchmark and predeclared model-level relation. The observed class was computed without using the expected label

Case	Controlled change	Expected class	Observed class
Identity	exact copy	strong equivalence	strong equivalence
Silent refinement	insert internal steps	weak equivalence	weak equivalence
Label mismatch	replace one observable family	not comparable	not comparable
Label order swap	preserve counts, change order	not comparable	not comparable
Added branch	preserve workflow, add outcome	reference simulated by candidate	reference simulated by candidate
Trace-equivalent branching	move point of choice	candidate simulated by reference	candidate simulated by reference

3.2. Independent mCRL2 oracle

To test implementation independence, each benchmark LTS was exported to the Aldebaran AUT interchange format and evaluated with `ltscompare` from mCRL2 202607.0 [4]. The external oracle computed strong bisimilarity, weak bisimilarity and exact weak-trace equivalence, treating tau as internal. These are separate executions of mCRL2 algorithms on the exported systems; the Python verdict is not supplied as input. Agreement is reported for strong and weak bisimulation, while exact weak-trace results are compared with the trace relation fixed by each construction. The current `ltscompare` interface does not provide a weak simulation preorder, so one-way weak-simulation classifications are not claimed to have an mCRL2 oracle. They are audited separately with a dual attacker–defender game. Instead of deleting invalid pairs from a candidate relation, the game grows the least set of positions from which an attacker can force an unmatched move. Every edge subset was enumerated for all one- and two-state LTSs over $\{a, \tau\}$: 260 systems and $260^2 = 67,600$ ordered comparison pairs. This is an exhaustive oracle for that declared finite universe, not a proof by testing for arbitrary model size.

3.3. Structural and linear-time comparators

Four transparent comparators isolate the information contributed by the formal relation. A structural score averages similarities in state count, edge count, observable-label multiset and out-degree multiset. Each count similarity is the smaller-to-larger ratio; multiset similarity is weighted Jaccard. The trace comparator declares a diagnostic match when $d_8 = 0$ and therefore records global order only to the selected depth, with no branching-time discrimination.

PN-GDDA was implemented directly from its published definition [27]. All connected induced directed bipartite graphlets on two through five nodes are generated algorithmically, giving 2, 6, 23 and 120 topologies by size (151 total). For each orbit slot j , every Petri-net node’s orbit degree is counted; arc

multiplicities, when present, multiply the occurrence count. If $D_G^j(k)$ is the number of nodes with orbit degree k , the normalised distribution and orbit agreement are

$$N_G^j(k) = \frac{D_G^j(k)/k}{\sum_{r>0} D_G^j(r)/r}, \quad A^j(G, H) = 1 - \frac{1}{\sqrt{2}} \sqrt{\sum_k \left(N_G^j(k) - N_H^j(k) \right)^2}. \quad (3.1)$$

PN-GDDA is the mean A^j from (3.1) over the 592 published slots. The six synthetic LTS pairs are presented to this Petri-net-specific comparator through an exact state-machine encoding: one place per state, one transition per LTS edge, one input and one output arc per transition, and one initial token. The five curated pairs are compared in their original Petri-net form. A score of at least 0.9 is a predeclared diagnostic decision for the binary benchmark; it is not treated as a formal equivalence relation.

The catalogue was audited independently against Holmes, the software used for the published method [23]. Both hash-recorded Holmes 1.1.1 and 2.0.1.2 contain 151 topologies and 592 orbit slots. A type- and direction-preserving automorphism reconstruction instead gives 576 distinct orbits; the discrepancy is confined to 12 five-node graphlets. The primary analysis preserves the Holmes 592-slot assignment so that the comparator is not silently redefined, while the 576-orbit partition is reported as a sensitivity analysis. A reproducible Java probe extracts all 151 topologies and all 592 root assignments from the hash-verified Holmes 1.1.1 JAR and requires entry-level agreement with the Python catalogue before evaluating the score. On a four-node $K_{2,2}$ input net and its one-arc deletion, the independent implementation reproduced the Holmes score (0.987202746587073 versus 0.9872027465870733), equal to 12 decimal places.

LTS-GDA is retained as a separate reachable-state comparator. On the simple undirected LTS projection it evaluates four graphlet orbits (edge, both three-node-path positions and triangle), then combines their mean agreement equally with weighted-Jaccard agreement on directed labelled edge, two-step walk, divergence and convergence motifs. Unlike PN-GDDA, it retains observable labels and can compare Petri nets, SBML-qual, GINML and inferred Boolean models through one representation. It uses the same predeclared 0.9 diagnostic threshold.

Binary equivalence ground truth is positive only for identity and silent refinement. Strong equivalence is included within weak equivalence for this binary evaluation; simulations are not counted as equivalence. To test whether the fixed 0.9 result is threshold-dependent, PN-GDDA accuracy is also evaluated at every distinct decision boundary induced by the six observed scores.

3.4. Scalability experiment

Linear workflows of 8, 16, 32, 64, 96 and 128 transitions were compared with an identical copy and a silently refined version. For each pair, strong bisimulation, weak bisimulation, both simulation directions and d_8 were measured separately. Wall-clock times are medians of three repetitions. The reference run used Python 3.11.8 on an Apple M3 Pro with 18 GiB memory. Runtime values are machine-dependent; state counts, edge counts, candidate relation sizes and classifications are deterministic.

3.5. Software and reproducibility

The formal engine, public-model parsers and benchmark use the Python standard library. Plotting and tabulation use NumPy, pandas, Matplotlib and NetworkX [11]; Petri-net artwork uses pydot and Graphviz. Independent checking uses the hash-pinned mCRL2 202607.0 release. The PN-GDDA audit records SHA-256 digests of both inspected Holmes JARs and emits its catalogue and numerical cross-check as machine-readable JSON. Submission figures are existing vector PDF outputs from the deterministic analysis pipeline; no figure or result was generated or modified by a generative-AI system. Fixed seeds control all synthetic modifications and interface permutations. The repository contains the exact model registry, benchmark generator, dual simulation-game oracle, unit tests, executed notebook, environment specifications, continuous-integration workflow and figure-generation script. Runtime CSV files are excluded from bit-for-bit checks because wall-clock measurements are hardware-dependent; all other generated results are deterministic.

4. Applications to qualitative biological network models

4.1. Controls from public logical models

Two externally authored models were obtained from the public GINsim repository [20]: the Boolean mammalian cell-cycle model of Traynard et al. [28], distributed as SBML-qual [6], and the multilevel p53–Mdm2 DNA-damage model associated with Abou-Jaoudé et al. [1], distributed as GINML. The exact downloaded files, source URLs and SHA-256 digests are retained in the repository. For the cell-cycle model, CycD=1 and all other components were initially zero. For the p53–Mdm2 model, the stored GINsim state damage was used.

Both rule sets were interpreted with unitary asynchronous semantics: in one transition, one enabled non-constant component moves one level toward its logical target. The observable label records the component and direction (component_up or component_down). Breadth-first enumeration then constructs the reachable LTS. For each public base model, three controls were predeclared: an exact copy must be strongly and weakly equivalent; inserting five internal steps must break strong but preserve weak and weak-trace equivalence; and renaming one reachable update family must break all three relations. These controls test parsing, state generation, AUT export and formal comparison on external dynamics. They do not establish that the published logical rules are complete biological representations.

As a semantic stress test, each public rule set was also executed synchronously: all unstable components were evaluated from the current state and moved one level in a single global transition. A sorted compound label recorded the component changes. Reachable-state overlap between the asynchronous and synchronous systems was reported, but no cross-semantics equivalence was assumed.

4.2. Blinded HPN-DREAM validation with held-out data

The data-backed validation uses the public HPN-DREAM phosphoproteomic time series for BT20, BT549 and MCF7 breast-cancer cell lines [15] and the corresponding CASPOTS Boolean-network families [24]. The archived families contain 72, 191 and 21 verified networks, respectively. Files were recovered from public commit 95e3c74 of the CASPOTS repository; the prior-knowledge network, learning sets, model families and mTOR-inhibitor test sets are retained with SHA-256 digests. Learning and test files remain separate throughout the workflow.

One representative per cell line was fixed before reading any test response. For each family, pairwise Jaccard similarity was computed between active Boolean clauses. The model with maximum mean within-family similarity was selected; exact ties were resolved lexicographically on the active-clause tuple using rational arithmetic. This structure-only rule selected zero-based rows 24, 137 and 9 for BT20, BT549 and MCF7. An anti-leakage unit test raises an exception if medoid selection attempts to open a test file.

The observational interface was the intersection of seven measured PI3K/MAPK/mTOR readouts. It comprised 4EBP1-pS65, AKT-pT308, BAD-pS112 and MAPK-pT202/Y204, together with MEK1-pS217/S221, mTOR-pS2448 and p70S6K-pT389. The perturbation set was fixed as the exact intersection of the three test panels: mTOR inhibition alone, IGF1 plus mTOR inhibition and 4EBP1 plus mTOR inhibition. Test values determined neither model selection nor interface membership. Each selected Boolean model was interpreted with unitary asynchronous semantics. Measured time-zero values initialized relevant readouts, other relevant nodes were zero, stimuli were clamped active and inhibitors inactive. Interface updates were labelled by component and direction; all other updates were τ . The resulting nine cross-cell pair-condition comparisons were evaluated by both implementations.

All nine comparisons were then repeated under a global synchronous stress semantics. Every unstable target was evaluated from the same source state and updated simultaneously. The transition label recorded the sorted set of changed interface readouts; a hidden-only global update was τ . This test asks whether the categorical relation survives a prominent alternative convention. It does not implement most-permissive semantics, and the asynchronous analysis remains primary because CASPOTS evaluates asynchronous trajectories [21, 24].

Experimental distance was the RMSE between the two cell-line profiles over common post-zero times and interface readouts. Time-zero measurements initialized the LTS but were excluded from this outcome to prevent data reuse. Formal classes were ordered from weak bisimulation through mutual and one-way simulation to non-comparability. Their association with experimental distance was evaluated with Spearman's ρ and an exact one-sided test that permuted the three pair labels within each condition ($3!^3 = 216$ permutations). Truncated trace distance was a secondary metric. One minus LTS-GDA similarity was evaluated by the same permutation scheme as a structural comparator; it was not used to select models, conditions or readouts.

Held-out compatibility was also recomputed with CASPOTS/Clingo. The network was fixed while the solver minimized the published weighted threshold-repair objective. Because streamed optimization models need not be final, the runner consumed the solve handle to completion and added deterministic secondary objectives for continuous squared error and source-row index. Clingo certified the final optimum, and each of 12 scores was reproduced twice identically. Each blind medoid was scored on all three test sets; selecting the best network from a whole family on its own test set was reported separately as a retrospective oracle, not a blind estimate. Native specificity was tested by exact sign permutation of the three native-versus-mean-cross differences. Setting every unmeasured relevant initial node to one was retained as a state-space stress test; its largest LTS had 27,648 states and was not entered into a quadratic exact cross-model relation.

4.3. Illustrative biological model set

The application contains five pairs of compact, safe Petri nets: DNA repair and genome instability (GIM), energy metabolism (DCE), cell-cycle control (SPS), cell death/autophagy (RCD), and an immune module (AID). Components were selected from published plant-human comparative studies [22, 8] and organized against the cancer-hallmark literature [12, 13]; dedicated reviews informed the qualitative representation of plant DNA-damage response, cell-cycle control and autophagy [29, 9, 17].

Places represent conditions or molecular states, transitions represent curated events, and the initial marking represents the declared starting context. A transition is observable only when its label belongs to the cross-system interface; species-specific intermediate steps are represented as τ . Transition-level provenance is stored in the repository and checked for complete name coverage. These nets are manually curated qualitative abstractions. They do not encode kinetics, concentrations, stoichiometric uncertainty or tissue context, and they were not inferred from independent experimental data.

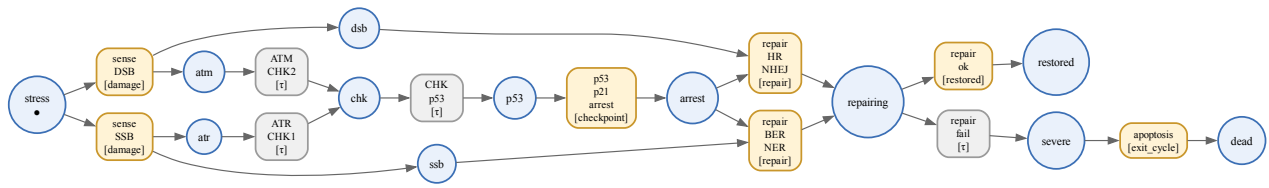
Figure 2 exposes the complete GIM encodings rather than only their computed verdict. Both nets implement damage sensing, checkpoint control, repair and restoration. Their terminal mechanisms differ: the animal/human net contains apoptosis, whereas the plant net contains an internal SMR step followed by differentiation or endoreduplication. The common label `exit_cycle` is a declared coarse observation, not evidence that those terminal mechanisms are biologically identical.

The interface is treated as a modelling hypothesis. Diagnostics include targeted single-label changes, random label permutations, seed sensitivity, a trace-depth sweep and silent-refinement invariance. The permutation values quantify sensitivity to scrambled interfaces inside these models; they are not biological significance tests and do not provide external validation.

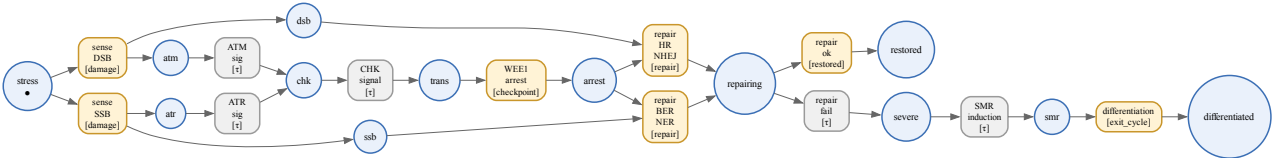
5. Results

5.1. The synthetic suite recovers every predeclared relation

The formal classification matched the construction-level class in all six cases (Table 3). Identity was strongly bisimilar. Inserting internal steps broke strong equivalence but preserved weak equivalence. Label mismatch and label-order change were both rejected. The added branch produced the intended one-way simulation, while the trace-equivalent branching pair was also separated by simulation direction rather than incorrectly promoted to equivalence.



(a) Animal/human GIM model



(b) Arabidopsis GIM model

Figure 2: Curated DNA-damage Petri nets used in the illustrative comparison: (a) animal/human and (b) Arabidopsis. Circles denote places and the initial token; rounded boxes denote transitions. Observable labels appear in brackets, while grey transitions labelled τ are internal. The diagrams are generated from the same executable definitions used by the analysis

Table 4: Binary equivalence accuracy on six construction-level cases

Method	Accuracy	False positive	False negative
Weak bisimulation	1.000	0	0
Trace equality ($k = 8$)	0.833	1	0
PN-GDDA-592 (≥ 0.9)	0.333	4	0
LTS-GDA (≥ 0.9)	0.667	1	1
Structural profile (≥ 0.9)	0.500	2	1

For the binary weak-equivalence task, weak bisimulation achieved 6/6 correct decisions. Trace equality achieved 5/6 because it produced a false positive on the branching-time case. Direct PN-GDDA achieved 2/6 and classified every pair as equivalent at the predeclared threshold. It assigned 1.0 to identity, silent refinement, label mismatch and label-order swap because the first two pairs share topology and the latter two modify labels that PN-GDDA does not encode. Added branch and trace-equivalent branching still scored 0.9676 and 0.9657. LTS-GDA achieved 4/6: silent refinement was a false negative because graphlets do not abstract τ , and the added local branch was a false positive at the predeclared threshold. The structural profile achieved 3/6, with false positives for label mismatch and label-order swap and a false negative for silent refinement (Table 4; Fig. 3). No PN-GDDA threshold attained more than 4/6: label mismatch and label-order swap both tied the two positive controls at 1.0, making those cases structurally inseparable by any score cutoff.

5.2. Independent software and public-model controls agree

The public cell-cycle model contained 13 Boolean components and generated 826 of 8,192 possible states with 3,413 reachable transitions. The four-component multilevel p53–Mdm2 model generated 17 of 36 possible states and 26 transitions (Table 5). Thus, the external controls exercise the method on substantially larger and differently encoded state spaces than the compact curated application.

All six public-model controls recovered their predeclared relations in both implementations (Table 6). Silent refinement added five states to each candidate without changing weak behaviour. The observable perturbations were rejected by strong bisimulation, weak bisimulation and exact weak-trace equivalence. Across these six controls and the six synthetic pairs, Python and mCRL2 agreed on 12/12 strong and

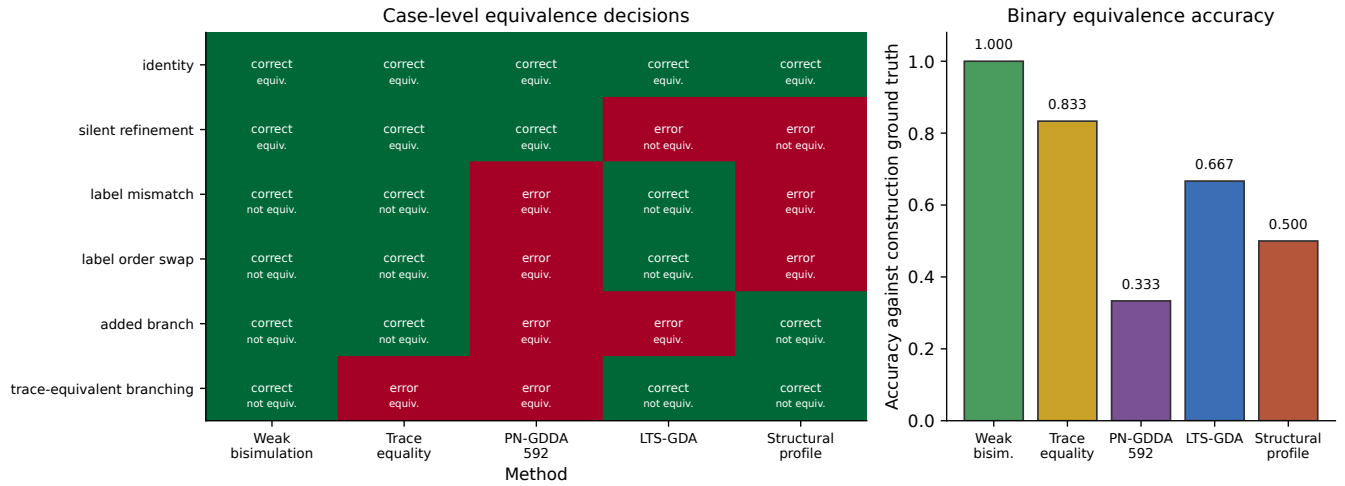


Figure 3: Independent benchmark. Left: correctness and predicted binary class for each case and method. Right: accuracy against the equivalence labels fixed by construction. The trace baseline misses the branching-time distinction; PN-GDDA is deliberately label-blind and calls all six encoded nets structurally similar, whereas labelled LTS-GDA improves on the summary profile but remains sensitive to silent refinement and local added structure

Table 5: Public logical models used for external workflow validation. Full space is the product of component level counts; only reachable states enter the formal comparison

Model	Format	Comp.	Full space	Reachable	Edges	Initial context
Mammalian cell cycle	SBML-qual	13	8,192	826	3,413	CycD=1; others zero
p53–Mdm2 DNA damage	GINML	4	36	17	26	stored state damage

12/12 weak decisions. The mCRL2 exact weak-trace result also matched all 12 predeclared trace controls, including acceptance of the trace-equivalent branching pair that both bisimulation implementations reject.

The independent attacker–defender oracle agreed with the production weak- simulation result on all 67,600 ordered comparisons among the 260 enumerated small LTSs. Alternative update semantics materially changed public reachability: the synchronous cell-cycle system contained 13 states, all shared with the 826 asynchronous states (Jaccard 0.0157), while synchronous p53–Mdm2 contained 8 states, all shared with the 17 asynchronous states (Jaccard 0.471). These audits support implementation correctness while showing that reachable behaviour is not semantics-neutral.

5.3. Held-out data support compatibility but not prognostic specificity

The structure-only representatives contained 23–29 active clauses (Table 7). Across the three shared perturbations, their reachable LTSs contained 2–44 states. None of the nine cross-cell comparisons was weakly bisimilar: six showed one-way simulation and three were not comparable. Python and mCRL2 agreed on every strong and weak decision (18/18), and mCRL2 rejected weak-trace equivalence in all nine pairs.

The primary ordinal class had a small positive association with between-cell experimental distance ($\rho = 0.091$), but the exact condition-stratified test was inconclusive ($p = 0.111$). Trace distance was negatively associated with the data distance ($\rho = -0.596$, $p = 0.556$), showing that the truncated linear-time index is not an empirical similarity surrogate in this small panel. LTS-GDA distance had the strongest observed association ($\rho = 0.669$), but its exact test also remained inconclusive ($p = 0.097$). It is therefore a useful comparator, not a validated response-distance predictor.

Under synchronous updating, four pairs had one-way simulation and five were not comparable. Seven of nine asynchronous classes were preserved. The BT20–BT549 pair under IGF1 plus mTOR inhibition

Table 6: Public-model controls. Entries list strong/weak/weak-trace outcomes; Python reports the first two relations and mCRL2 reports all three. Y=yes and N=no

Model	Control	States ref./cand.	Edges ref./cand.	Expected S/W/T	Python S/W	mCRL2 S/W/T
Cell cycle	exact copy	826/826	3,413/3,413	Y/Y/Y	Y/Y	Y/Y/Y
Cell cycle	silent refinement	826/831	3,413/3,418	N/Y/Y	N/Y	N/Y/Y
Cell cycle	observable perturbation	826/826	3,413/3,413	N/N/N	N/N	N/N/N
p53-Mdm2	exact copy	17/17	26/26	Y/Y/Y	Y/Y	Y/Y/Y
p53-Mdm2	silent refinement	17/22	26/31	N/Y/Y	N/Y	N/Y/Y
p53-Mdm2	observable perturbation	17/17	26/26	N/N/N	N/N	N/N/N

Table 7: HPN-DREAM model families and blinded structure-only representatives

Cell line	Family	Medoid row ^a	Clauses	Mean Jaccard
BT20	72	24	29	0.815
BT549	191	137	25	0.854
MCF7	21	9	23	0.851

^a Zero-based row; selected from model structure only.

and the BT20–MCF7 pair under 4EBP1 plus mTOR inhibition changed from one-way simulation to non-comparability. Thus the broad absence of equivalence persisted, but two directional-containment statements were update-semantics dependent.

CASPOTS reproduced compatibility with the held-out test data. Native medoid RMSE was 0.33024 for BT20, 0.31335 for BT549 and 0.27726 for MCF7; corresponding discretization RMSE was 0.32944, 0.30079 and 0.27726. The native medoid was best or tied for BT20 and MCF7, but not for BT549 (Fig. 4a). Across the three targets, the native excess-RMSE advantage over the mean of both cross-cell medoids was 0.00212 and was not significant under the exact paired sign test ($p = 0.25$). The independently selected formal class likewise did not significantly order experimental distances (Fig. 4b). These results validate held-out compatibility and deterministic reproduction of the published scoring logic, while directly rejecting a claim of demonstrated cell-specific predictive discrimination.

5.4. Runtime grows with the candidate relation

Across the two scaling series, complete runtime rose from approximately 0.4 ms for 9-state reference models to approximately 0.6 s for the largest pairs. The 128-step identity pair contained $129 \times 129 = 16,641$ candidate state pairs; silent refinement increased the candidate to 145 states and the relation to 18,705 pairs. Both series retained their expected classifications at every size (Fig. 5). Two simulation computations accounted for the largest runtime share. These measurements establish feasibility for compact and medium qualitative modules, not for genome-scale state spaces.

5.5. Case-study outputs are conditional relations between curated models

The ten curated Petri nets generated reachable systems of 5–14 states. Under the declared interfaces, the GIM, DCE and SPS model pairs were weakly, but not strongly, bisimilar and had $d_6 = 0$. The RCD plant model was simulated by the animal model but not conversely ($d_6 = 0.143$). The AID pair satisfied neither simulation direction ($d_6 = 0.429$; Table 8).

Direct PN-GDDA reached the same binary decision for all five native pairs despite their different behavioural classes (Table 9). The RCD nets received a perfect structural score although only one simulation direction held; AID scored 0.9907 although neither direction held. Replacing the 592 Holmes slots with the 576 automorphism-distinct orbits changed no score by as much as 0.001 and changed no threshold decision. Thus, the disagreement is not caused by the catalog audit: local unlabelled Petri-net structure and reachable labelled behaviour answer different questions.

In GIM, apoptosis is the animal terminal fate, whereas differentiation and endoreduplication are plant terminal fates. All three outcomes are represented through the common coarse observable `exit_cycle`.

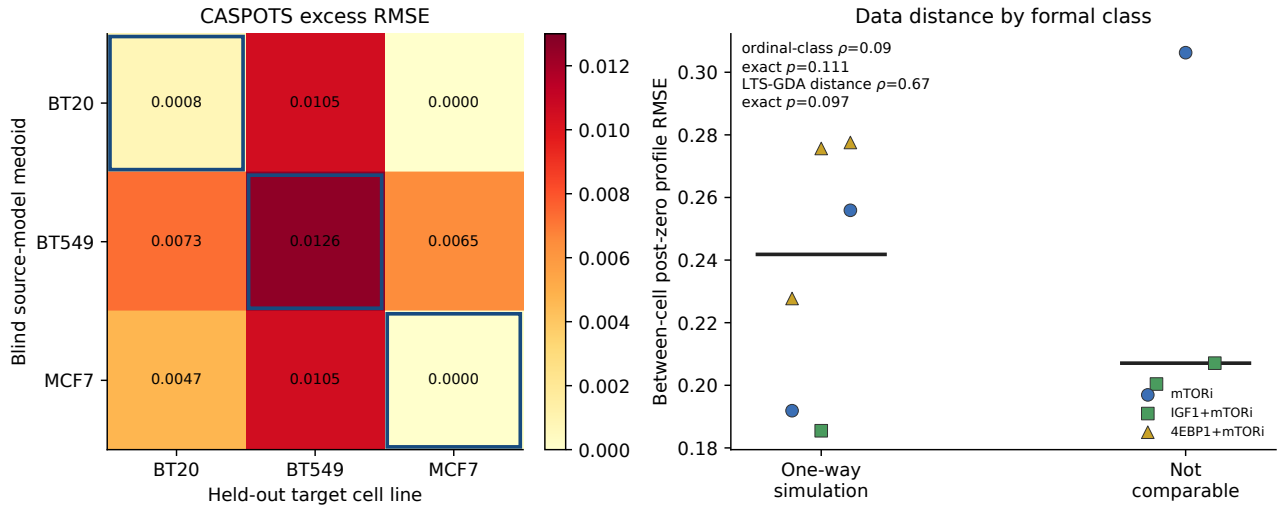


Figure 4: Blinded HPN-DREAM validation. (a) CASPOTS excess RMSE above the discretization reference for every structure-only source medoid and held-out target cell line; blue boxes mark native source–target combinations. (b) Between-cell profile RMSE under the primary formal class for three shared perturbations. Horizontal lines are class medians. The exact test permutes cell pairs within perturbation; the annotation reports both ordinal-class and LTS-GDA-distance associations. Neither panel supports robust native-cell specificity

Table 8: Relations between the curated case-study model pairs. These are model-level, interface-conditional outputs; P denotes the plant model and A the animal model

Module	Encoded process	Strong	Weak	$P \preceq A$	$A \preceq P$	d_6
GIM	DNA repair	no	yes	yes	yes	0.000
DCE	energy metabolism	no	yes	yes	yes	0.000
SPS	cell-cycle control	no	yes	yes	yes	0.000
RCD	cell death/autophagy	no	no	yes	no	0.143
AID	immune control	no	no	no	no	0.429

The weak-equivalence result therefore says that the two encoded models agree at this selected resolution. A finer interface that distinguishes those outcomes would change the classification. This example makes the conditional nature of the method explicit rather than treating the interface as a neutral measurement device.

The corresponding reachability systems in Fig. 6 make the weak relation inspectable. Observable paths for damage, checkpoint, repair, restoration and cycle exit can be matched, while additional internal paths are absorbed by τ closure. The figure therefore connects the Petri-net construction to the state-pair relation used by the algorithm.

The RCD pair supplies a contrasting one-way result (Fig. 7). Both encodings include stress, mTOR inhibition, autophagy and survival. The animal model additionally exposes canonical caspase apoptosis, whereas the plant model proceeds through an internal metacaspase event. Consequently, the plant behaviour is simulated by the animal model under the declared interface, but the reverse direction fails. This is a containment statement about the two encodings, not a claim that either net exhausts cell-death biology.

Randomly subdividing internal structure preserved every categorical result, as required by weak semantics. Renaming individual observables changed the expected model relations, confirming that interface labels affect the computed result. Label-permutation diagnostics separated the three weakly equivalent curated pairs from scrambled versions, but these tests only assess the internal specificity of the selected interface. They cannot establish biological validity because the original labels and networks were manually curated (Fig. 8).

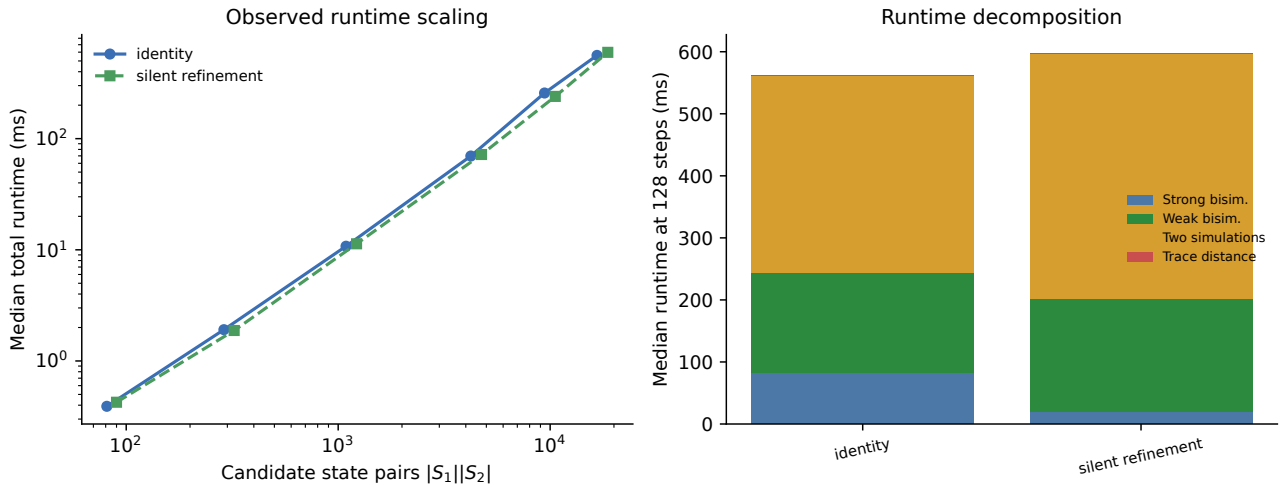


Figure 5: Scalability on deterministic workflows. Left: median total runtime against candidate state-pair count on logarithmic axes. Right: decomposition of the largest run into strong bisimulation, weak bisimulation, two simulations and trace distance. Timings are machine-dependent; model sizes and outcomes are deterministic

Table 9: Direct PN-GDDA on the original curated Petri-net pairs. “592” is the published/Holmes slot catalog; “576” is the automorphism-partition sensitivity. The ≥ 0.9 column is a diagnostic threshold, not a formal relation

Module	PN-GDDA-592	Orbit-576	≥ 0.9	Behavioural relation
GIM	0.9976	0.9975	yes	weak bisimulation
DCE	0.9969	0.9968	yes	weak bisimulation
SPS	1.0000	1.0000	yes	weak bisimulation
RCD	1.0000	1.0000	yes	one-way simulation
AID	0.9907	0.9904	yes	not comparable

6. Discussion

6.1. Semantic meaning of the graded comparison

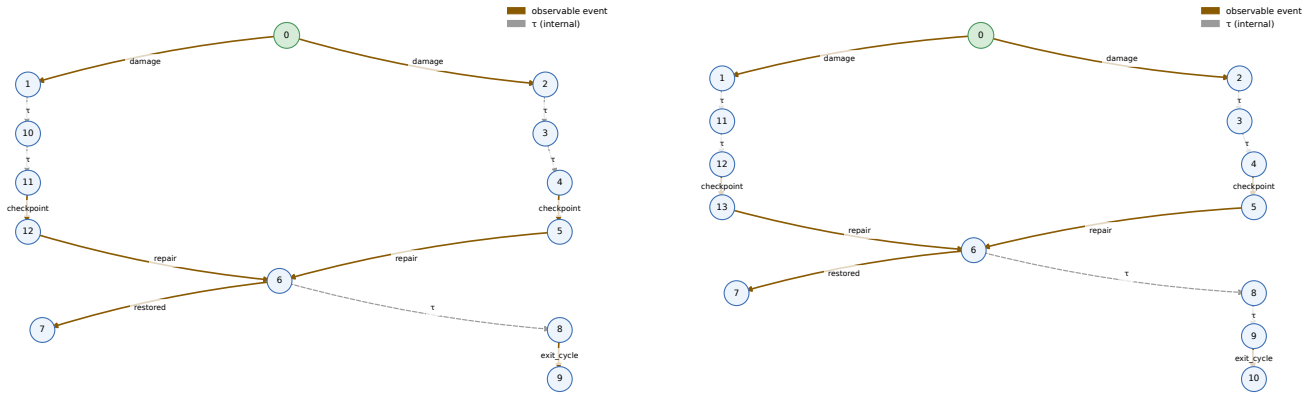
The classifier turns an ordered pair of executable models into one interface-conditional statement. Its priority is supported by Proposition 2.11: a strong result already entails the corresponding weak and simulation results, while a weak result already entails both simulation directions. Reporting only the strongest satisfied class prevents the same pair from receiving several apparently competing labels. The one-way classes preserve orientation and therefore express containment rather than equivalence.

The fixed-point result is about the algorithm that was executed, not an idealized variant. Theorem 2.9 covers the in-place deletion order, the implemented treatment of τ , and termination on finite state spaces. It also separates mathematical correctness from software validation: the proof applies to arbitrary finite inputs under the stated semantics, whereas tests can expose implementation discrepancies only on the inputs that they exercise.

6.2. Structure, traces and branching are non-interchangeable

Proposition 2.13 gives a general reason why identical unlabelled structure cannot determine labelled behaviour. The controlled label mismatch and order swap instantiate that limitation: both receive a PN-GDDA score of 1.0 although neither simulation direction holds. Example 2.14 supplies the complementary linear-time limitation. Its two systems have the same exact finite traces, yet they differ in the point at which choice is resolved and satisfy only one simulation direction.

Direct PN-GDDA achieved 2/6 on the predeclared binary task, but this is not a software failure or a claim that PN-GDDA is unsuitable for its intended purpose. The independent Holmes score agrees to 12 decimal places, and the 592- and 576-slot catalogues give the same threshold decisions. Transition



(a) Animal/human reachability system

(b) Arabidopsis reachability system

Figure 6: Reachability systems generated from the GIM Petri nets: (a) animal/human and (b) Arabidopsis. Numbered nodes are reachable markings, the initial node is distinguished by its fill, solid labelled arcs are observable moves and dashed arcs are internal τ moves

labels, initial marking and reachable branching lie outside the structural statistic by design. LTS-GDA recovers labelled local motifs and improves to 4/6, but it still does not implement silent abstraction or recursive successor matching. Its held-out association is the strongest observed ($p = 0.669$), while the exact $p = 0.097$ remains inconclusive. The appropriate conclusion is to report structural, trace and branching views as answers to different questions.

6.3. Observational interfaces and abstraction

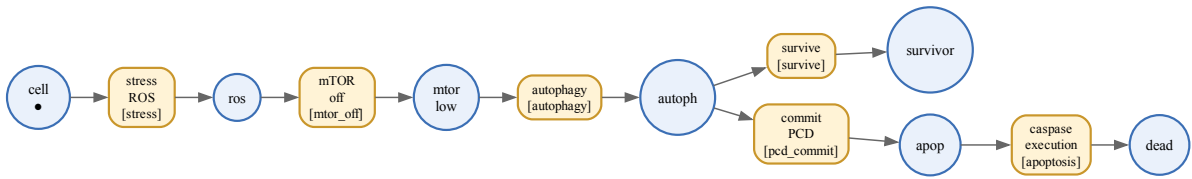
The relation is indexed by Σ . Relabelling an event as internal changes which paths are abstracted, and merging distinct outcomes under one label can coarsen the result. In the GIM application, apoptosis and differentiation or endoreduplication share the declared observable `exit_cycle`. Weak bisimilarity therefore describes agreement only at that selected resolution. It is not a statement that the terminal mechanisms are biologically identical.

Proposition 2.12 justifies the controlled subdivisions used in the robustness analysis. It does not license arbitrary Petri-net refinement: added conflicts, interleavings or observable choices may change the reachable LTS. Update semantics is another part of the model contract. The asynchronous-synchronous GINsim differences and the two changed HPN classes show that a relation should not be described as semantics-neutral. Most-permissive, general-asynchronous and stochastic conventions were not evaluated [21].

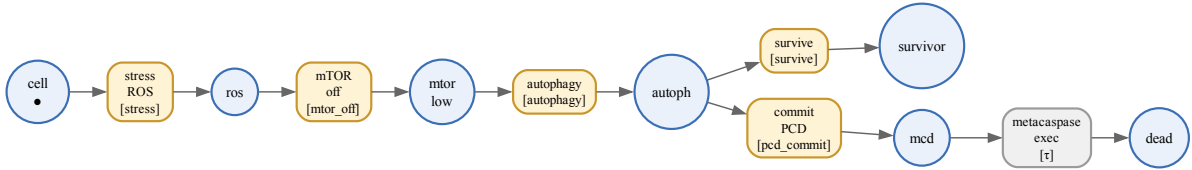
6.4. Independent computational evidence

The six synthetic pairs isolate known relations before any biological case is examined. The formal classifier recovers all six, while the comparator errors occur in the cases predicted by their information loss. Across the six synthetic pairs, six public-model controls and nine HPN comparisons, Python and mCRL2 agree on all 42 reported strong/weak decisions. mCRL2 additionally confirms the exact weak-trace controls, including acceptance of the branching pair that both bisimulation implementations reject.

The dual attacker-defender implementation checks the simulation orientation not available through the used `ltscompare` interface. Its agreement on all 67,600 ordered pairs is exhaustive only within the declared universe of 260 one- and two-state LTSs over $\{\alpha, \tau\}$. It is evidence against small-system implementation defects, not a proof for arbitrary systems. The transformed GINsim controls similarly test parsing, execution and export, but their expected relations follow from controlled transformations. Distinct HPN model families and held-out measurements provide a separate empirical check rather than a replacement for formal validation.

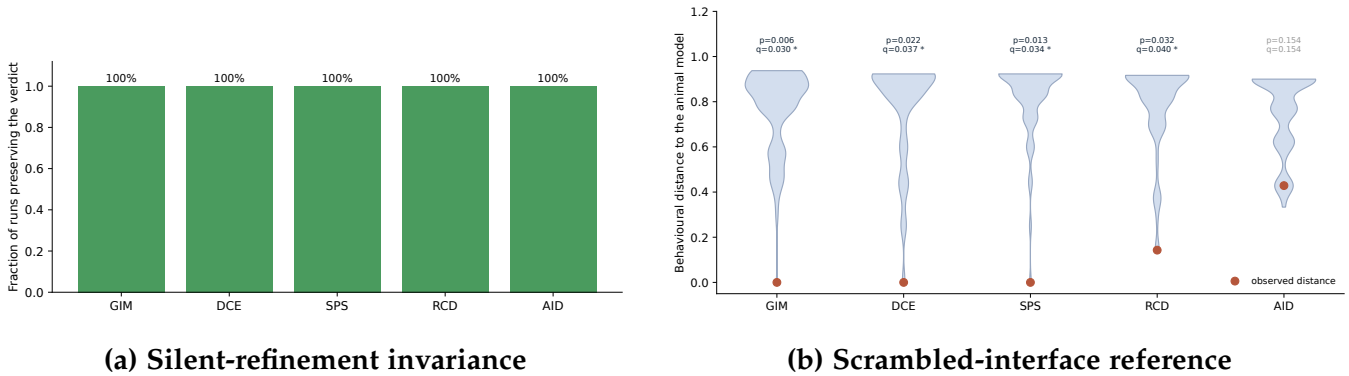


(a) Animal RCD model with an observable caspase-apoptosis branch



(b) Arabidopsis RCD model with an internal metacaspase branch

Figure 7: Curated cell-death/autophagy Petri nets illustrating the source of the one-way simulation result: (a) animal and (b) Arabidopsis. Symbols and transition styling follow Fig. 2



(a) Silent-refinement invariance

(b) Scrambled-interface reference

Figure 8: Internal robustness diagnostics for the curated case study: (a) verdict preservation under 300 silent refinements per module and (b) observed distance against 2,000 scrambled-interface distances. Bars and numeric labels report refinement invariance; violin distributions show the label-permutation reference and points show observed distances. Reported p- and Benjamini–Hochberg q-values quantify model-internal interface specificity and are not experimental biological significance tests

6.5. Biological applications and empirical limits

The HPN-DREAM application has nine pair-condition observations from three cell lines. Six comparisons show one-way simulation and three show non-comparability; none is weakly bisimilar. The ordinal formal class has $\rho = 0.091$ with experimental distance and an inconclusive exact $p = 0.111$. Truncated trace distance also lacks supportive concordance. CASPOTS reproduces held-out compatibility, but the native-versus-cross advantage is not established ($p = 0.25$). CASPOTS RMSE is a minimum-distance compatibility score over admissible trajectories, not a unique time-series prediction [24]. These results do not support prognostic use or cell-line specificity.

The curated Petri-net outputs are also conditional. The models omit kinetics, quantities, noise, stoichiometric uncertainty, tissue context and independent experimental inference. Different curators could select different arcs, initial markings or observables. Provenance coverage documents the choices but does not make them independent evidence. Label-permutation values measure sensitivity inside the constructed models and are not biological significance tests. In particular, weak relations between an Arabidopsis net and an animal net do not establish organism-level functional equivalence.

6.6. Computational limitations and use

The implementation enumerates states explicitly and stores up to $|S_1||S_2|$ candidate pairs. Silent closures are cached, but visible weak target sets are reconstructed on demand. The scaling study reaches 145 states and 18,705 candidate pairs; it does not establish genome-scale performance. Symbolic state representations, partition refinement and compositional checking are appropriate directions for larger systems.

PN-GDDA is evaluated directly on the native curated nets and canonical state-machine encodings of the six controlled LTS pairs. Applying it to SBML-qual or inferred Boolean rules would require a non-unique reverse translation into Petri-net structure. The difference between 592 published/Holmes slots and 576 automorphism-distinct orbits also remains a catalogue question beyond the compact sensitivity analysis. Future evaluations should use larger repository Petri nets, prospectively expert-labelled relations, biological replicates, additional update semantics and blinded interfaces. In its current form, the framework is most appropriate for auditing two explicit qualitative models at a predeclared observational resolution.

7. Conclusion

We formulated a graded comparison framework for finite labelled Petri nets and other executable qualitative models under a declared interface. The classifier reports the strongest supported relation among strong bisimilarity, weak bisimilarity, mutual or one-way weak simulation and non-comparability. Its priority follows from the relation hierarchy, and the implemented deletion algorithm computes the corresponding greatest fixed points and terminates on finite LTSs. Controlled silent refinement, label relabelling and the trace-equivalent branching example establish why internal abstraction, unlabelled structure, linear traces and branching cannot be treated as interchangeable.

Construction-level cases, mCRL2 and the independent simulation game provide separate computational checks, including complete agreement on 67,600 ordered pairs within the declared small-LTS universe. Applications to GINsim, HPN-DREAM/CASPOTS and curated Petri nets demonstrate cross-format execution and make interface and update-semantics dependence visible. They do not establish predictive performance, cell-line specificity or organism-level equivalence. The resulting contribution is a reproducible formal model audit whose mathematical outputs and empirical diagnostics remain explicitly separated.

Data and code availability

Curated definitions, hash-verified public GINsim models, HPN-DREAM/CASPOTS model families, normalized perturbation-response files, benchmark metadata and generated result tables are available at <https://github.com/cero1979/BisimulationMol>. Source code is released under the MIT License. The repository documents separate commands for external-model validation, HPN-DREAM validation, the Holmes PN-GDDA check, figure regeneration, unit tests and notebook execution. No confidential or participant-level data were used.

Funding

The author received no specific funding for this work.

Competing interests

The author declares no financial or non-financial interests directly or indirectly related to this work.

Ethics statement

Not applicable. The study uses computational models and previously published aggregate information; no human participants, human samples or live animals were involved.

Author contributions

Carlos Ramirez Ovalle: conceptualization, methodology, software, formal analysis, validation and visualization; writing—original draft; writing—review and editing.

Declaration of AI Use

During the preparation of this work, the author used OpenAI Codex for the following purposes: improving language and readability, checking grammar, formatting references, adapting the LaTeX source to the journal template, and assisting with reproducibility checks. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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